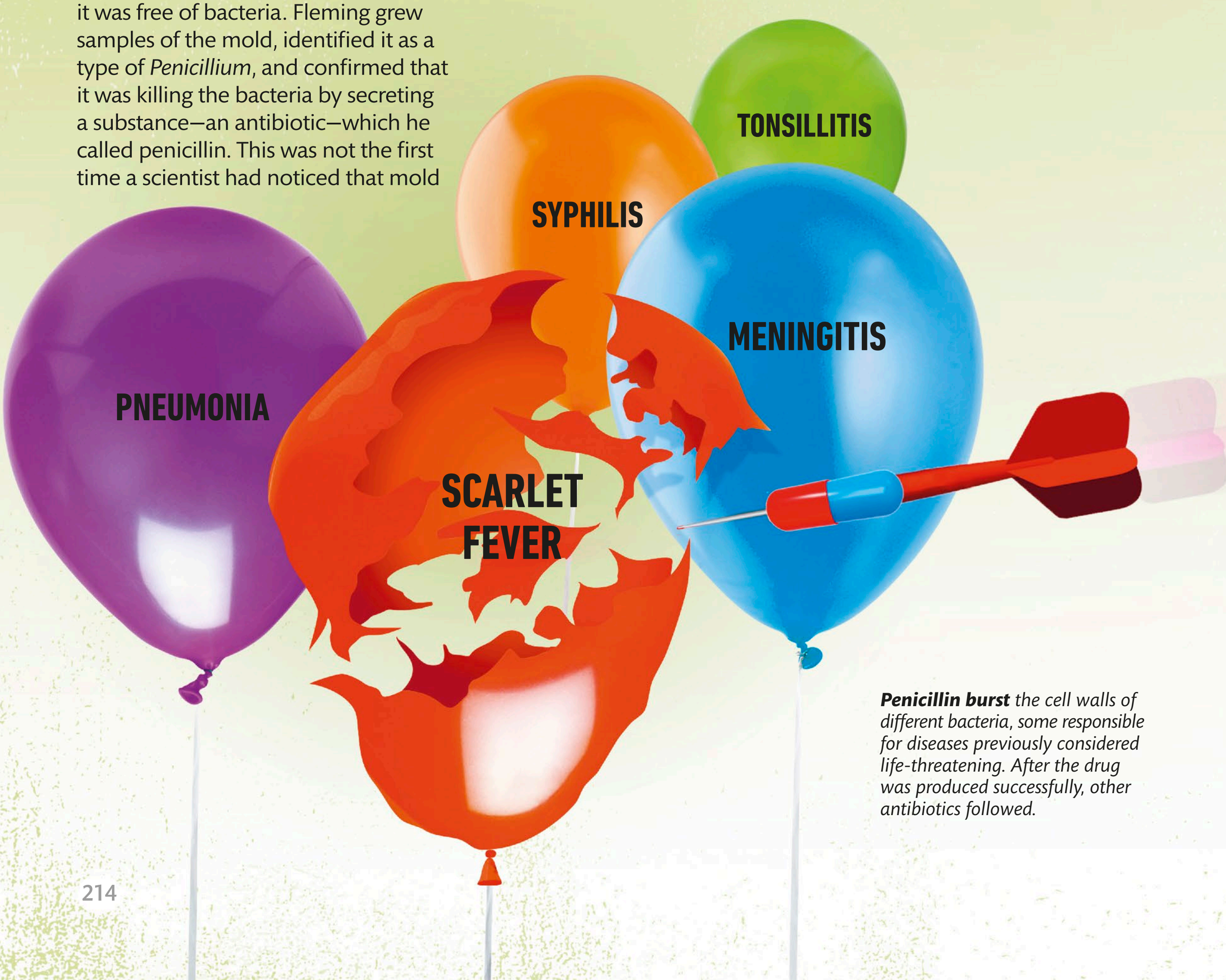
On September 3, 1928, a chance discovery changed everything. Fleming had been on vacation for weeks, and on his return, he started to clean up his notoriously messy laboratory. He was studying the bacterium *Staphylococcus*, the strains of which cause many diseases. Petri dishes containing the bacterium were stacked in the sink, waiting to be cleaned.

An accidental find

Fleming noticed something surprising. One of the dishes had been infected by some sort of mold, and the area around it was free of bacteria. Fleming grew samples of the mold, identified it as a type of *Penicillium*, and confirmed that it was killing the bacteria by secreting a substance—an antibiotic—which he called penicillin. This was not the first time a scientist had noticed that mold

"For the birth of something new, **there has to be a happening.**"

Alexander Fleming



Penicillin burst *the cell walls of different bacteria, some responsible for diseases previously considered life-threatening. After the drug was produced successfully, other antibiotics followed.*